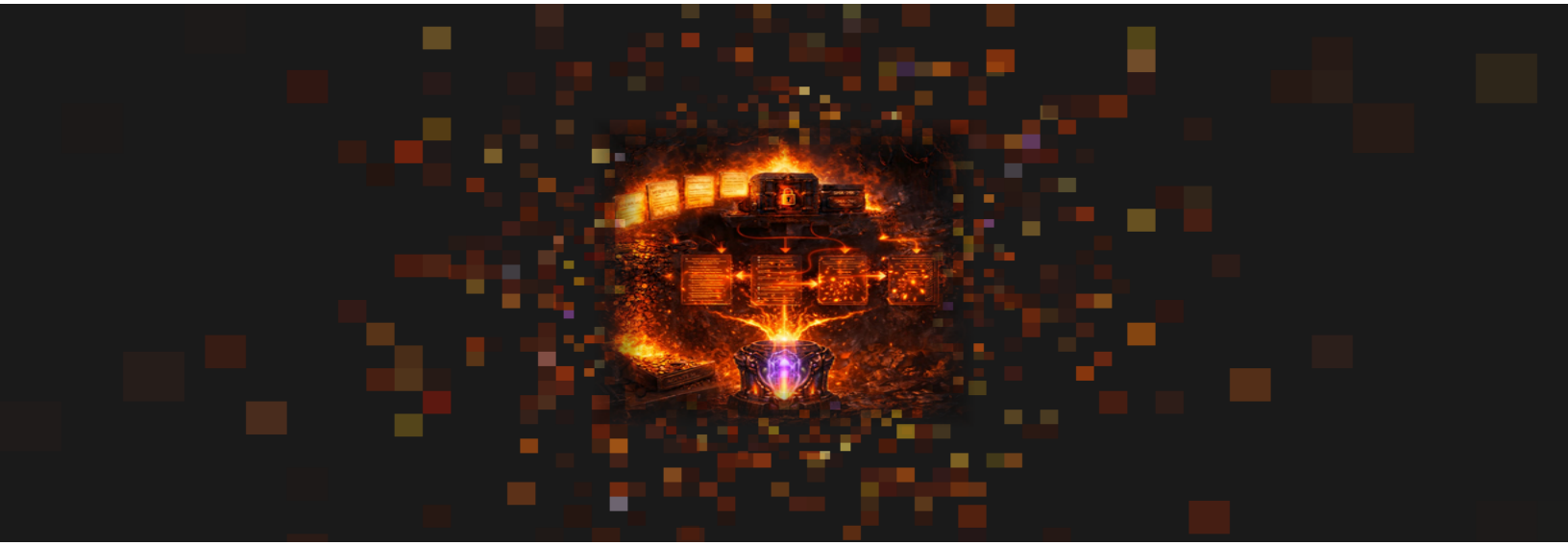

Stop paying for tokens your AI never needed to read.

THE PROBLEM

Your AI re-reads everything, every time

When you use an AI assistant through a framework like OpenClaw, something expensive happens behind the scenes. Every time you send a message, the system resends your entire set of workspace files to the AI. Not once at the start. *Every single message.*

In a 50-message conversation, that means the AI processes over 350,000 tokens of the same static files before it even looks at what you actually asked. Most of that content has nothing to do with your question.

> *Imagine going to a library and being forced to re-read every book on the shelf before the librarian answers your question. That's what your AI does right now. Every. Single. Time.*

THE SOLUTION

Send only what matters

SMELT (Schema-Aware Markdown Compilation for Efficient Local Token Inference) is a compiler that takes your workspace files and creates a denser version that preserves all the information but uses far fewer tokens. When you ask a question, it sends only the parts relevant to that specific question.

Think of it like a librarian who knows every book on the shelf, and when you ask a question, pulls out just the right page instead of handing you the entire library.

THE NUMBERS

Real results on real hardware

95%	6%	73
fewer tokens per query	faster startup time	tokens instead of 1,373

> *A simple question like "Who is Kane?" normally forces the AI to process 1,373 tokens of workspace files. With SMELT, it processes just 73. Same answer, 95% less work.*

WHO BENEFITS

Three audiences, three wins

- **API users (you pay per token):** Every unnecessary token is money. SMELT cuts workspace token cost by up to 95% on every call.
- **Local model users:** Every token is time. Fewer tokens means faster responses. 6% faster startup measured on Apple M3-Ultra.
- **Providers (Anthropic, OpenAI):** Millions of users, billions of calls, each carrying thousands of unnecessary tokens. Infrastructure waste at planetary scale.

HOW IT WORKS

Four layers, one idea

The idea is borrowed from how computers have always worked: you write code in a human-readable language, then a compiler turns it into something efficient. SMELT does the same for AI workspace files.

1. **Archive**
   Your original files are saved exactly as they are, verified with a digital fingerprint.
2. **Compile**
   Readable markdown becomes a denser format. Headings become short codes, formatting is stripped, but every fact, name, and relationship is preserved.
3. **Compress**
   Common phrases are replaced with short tokens, like a dictionary that makes everything compact.
4. **Select**
   The big one. When you ask a question, SMELT scores every record and sends only what's relevant. This is where the 95% savings come from.

> *One surprising finding: converting workspace files to JSON (a common "optimization" trick) actually makes things 30% worse. SMELT works because it understands the structure, not just the text.*

HOW THIS CAME TO BE

Built by a builder, for builders

I built SMELT because I was running a local AI model on my Apple M3-Ultra and I could literally watch it waste time processing the same files over and over. Every message, thousands of tokens of static content re-read before my actual question got touched. It felt like watching a car idle in traffic, burning fuel going nowhere.

So I built a compiler. The same idea that has made software fast since the 1950s: don't interpret source code at runtime, compile it first.

I built it with the help of GPT, Claude, and Codex. They helped design the architecture, run the benchmarks, review the paper, and make everything better. They're credited because they deserve to be.

THE COMPANION

SMELT + RUNE

SMELT has a companion project called RUNE, which solves a different but related problem: your AI forgets who you are between sessions. RUNE saves your identity and conversation history to an encrypted file you carry with you. SMELT is planned as the retrieval engine that makes RUNE's memory search fast and efficient.

Together, they solve the two biggest frustrations in AI: wasted tokens and lost relationships.

> *SMELT makes your AI faster and cheaper. RUNE makes your AI remember you. Both are free for personal use and open source.*

GET STARTED

SMELT is a Python tool you can run today. The compiler, the paper, and full benchmark results are all available on GitHub.

GitHub	github.com/TooCas/SMELT
Paper	doi.org/10.5281/zenodo.19380983
RUNE	github.com/TooCas/RUNE

Designed by Edmund Lister and shared freely for personal use
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